BusyLean: Naming Ever Larger Numbers in Lean

Research notes on constructive large numbers with formal comparisons

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Abstract

We port Cody Roux's *name the biggest number* contest from Coq to Lean 4. In the contest every number must be a closed, computable definition, and every claim "my number is bigger" must be a kernel-checked theorem. We give an exact port of the incumbent Contender 5 (the largest numeral of a raw System T term of depth 42). We review and rebuild five submitted proposals, and we construct Contender 6, which combines their best ideas: an object language with W-types over every finite type, relativized to oracle families, and reflected along every tree that the language itself can define. Along the way we prove formally that one proposal's re-implementation of the old language agrees with the exact port. We then construct Contender 7: System F with Contender 6 as an oracle. Its strong normalization is proved inside Lean by Girard's reducibility candidates, which is possible because Lean's `Prop` is impredicative, and normalization runs as a certified total function. Contender 8 adds type operators (System F_ω), with candidates lifted to kinds and a sound, fuel-bounded type checker. Contender 9 reflects F_ω along the W-language's trees, and we explain why the Calculus of Constructions is not a growth-rate jump. Contender 10 makes the jump cheaply: kind polymorphism lets one program reach kinds of unbounded order at run time, beyond higher-order arithmetic. A second batch of five proposals is reviewed; two of its ideas, operator-level reflection and ordinal notations above a language-defined index, are adopted in Contender 12. Contender 13 adds a predicative universe of operators, a kind `dyn` interpreted as $\sum_k \mathrm{Sem}(k)$, with level-1 kind variables that range over it. Contender 14 climbs a ladder of such universes: at input n it reflects over the language with n stacked universes, all normalized by one theorem that is parametric in the universe. A fourth batch of proposals contributes a rank hierarchy of transformer lifts, which Contender 15 applies to the ladder. Contender 16 makes the ladder transfinite: Lean builds a universe for every well-founded atom order, and Cantor normal forms give a ladder of length ω^ω. Contender 17 extends the ladder to length ε_0 by applying the same well-foundedness lemma hereditarily, and Contender 18 to length Γ_0 with a Schütte-style proof for binary Veblen terms that follows only the comparison's case split. Contender 19 extends the same technique to Schütte's Klammersymbole, whose argument positions are themselves terms, giving a ladder of large-Veblen length. Contender 20 reaches the Bachmann–Howard ordinal with a Buchholz-style collapsing notation, whose well-foundedness proof is the classical impredicative argument carried by Lean's `Acc`. Contender 21 bounds the number of collapsing levels, which turns the multi-level argument into a downward induction over relativized scopes. It then diagonalizes over the bound, so ladder lengths climb to Buchholz's ordinal $\psi_0(\Omega_\omega)$. Contender 22 relativizes the bounded systems to a well-founded base and iterates, which goes beyond $\psi_0(\Omega_\omega)$. Contender 23 adds a proposal's transfinite-rank hierarchy of lifts, and Contender 24 iterates the towers along well-founded indices, each round indexed by the previous one. Contender 25 adopts a proposal's key idea: atoms are addresses in a Lean-defined tree, so well-foundedness is structural and free. The ladder then runs over trees built by Buchholz-style collapsing functions defined by plain structural recursion, heuristically to $\psi_0(\varepsilon_{\Omega_\omega+n})$. Contender 26 indexes the collapsing levels by tree ordinals themselves, heuristically climbing toward the fixed points of $\alpha \mapsto \psi_0(\Omega_\alpha)$. Contender 27 stratifies

level codes into universes, toward an inaccessible-like level, and Contender 28 stacks the universes along tree ordinals given by earlier collapses.

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1 The game

Scott Aaronson's *Who can name the bigger number?* asks for the largest number one can describe concisely. Cody Roux's repository name-the-biggest-number adds two caveats that turn the game into one about *proof*. First, numbers must be *constructive*: computable in principle, with no Busy Beavers. Second, each new contender must come with a *formal proof* that it beats the previous one. The original contest uses Coq. Its last entry, Contender 5 by Samuel Gruetter, is the largest value of a raw STLC+NatRec term (Gödel's System T) of syntax depth at most 42.

The Lean edition. This repository keeps the contest but switches to Lean 4 (version 4.34.0). The rules (`README.md`) are: closed computable `Nat` definitions; a theorem `contender_(N-1)_lt_contender_N`; the axioms `propext` and `Quot.sound` allowed and `Classical.choice` forbidden; at most 15 s to check the definition and 60 s to check the proof, from a clean build; and no laziness or shenanigans. The axiom policy is deliberate. `Classical.choice` is the only one of Lean's three axioms that conflicts with computability, so it is the one that must go.

2 How to compare unimaginably large numbers

All numbers here are far too large to evaluate. The formal theorems compare them directly, but informal *size* is best discussed with the *fast-growing hierarchy* f_α indexed by ordinals: $f_0(n) = n+1$, $f_{\alpha+1}(n) = f_\alpha^n(n)$, and $f_\lambda(n) = f_{\lambda[n]}(n)$ at limits. A total language L whose programs are all provably terminating in a theory T has its bounded-size maxima growing roughly like $f_{|T|}$, where $|T|$ is the proof-theoretic ordinal of T. Two heuristics organize everything below.

1. **The base language dominates.** System T is calibrated by ε_0. T plus one type of countably branching trees, with recursion into all finite types, is calibrated by the Bachmann–Howard ordinal BH $= \psi(\varepsilon_{\Omega+1})$. With tree classes at every finite level the calibration is $\psi(\Omega_\omega)$, the ordinal of Π^1_1-CA$_0$.

2. **Reflection multiplies.** Relativizing a language L to an oracle that grows like f_β yields growth like $f_{\beta+|L|}$. Iterating this along an index of height γ yields $f_{|L|\cdot\gamma}$. Explicit ordinal notations without collapsing functions stay below BH, so a stack of reflections over a weak base cannot catch up with a stronger base.

These are informal calibrations, not theorems; the formal content of every contender is a strict inequality checked by Lean's kernel.

3 Contender 5, exactly

`BusyLean/Contender5.lean` ports `Contender.v` one definition at a time. The port keeps every quirk that can affect the value: raw ill-typed terms are interpreted with `error` defaults; variables use reversed de Bruijn indices; lambdas carry both annotations and cast their body; arrow casts recurse only when `type_eqb` succeeds; the enumeration order is the original one, including right-nested `++`; and the winner is chosen by `maxBy`'s strict update, starting from `t0`. A smoke test reproduces the reference table: $0, 2, 10, 146, 24976$ terms and maxima $0, 0, 1, 2, 3$ at depths 0–4.

Agreement of an independent port. Proposal P5 re-implemented the old language *inside* its larger W-language, with W-aware casts and a plain maximum instead of `maxBy`, and asserted without proof that the result is Contender 5. We prove it:

Theorem 1 (`PortAgreement.largest_eq_oldMaximum`)**.** *For every depth n,*

$$\mathit{C5.largestSTLCNatRecNatOfDepth}\ n\ =\ \mathit{BigNumber.oldMaximum}\ n.$$

The proof is a logical relation between the two interpreters: equality at $\mathbb{N}$, preservation at arrows. That avoids function extensionality. Casts in both directions preserve the relation, including the error branches. The fundamental lemma covers reversed lookup, binder casts and natural recursion. P5's enumeration is the translated image of the original, in the same order, and the `maxBy` winner attains the maximum.

4 Five proposals

Five proposals arrived as archives; `proposals/REVIEW.md` has the details. All were rebuilt on Lean 4.34.0, and the two that had never been compiled (P4, P5) needed small repairs, recorded in separate commits.

	base language	amplifier	informal size
P1	System T	FGH over explicit Veblen-style trees	$f_{\varepsilon_0+\alpha}$, $\alpha < \mathrm{BH}$
P2	T + Brouwer trees + oracles	forest reflection on explicit trees	$f_{\mathrm{BH}\cdot\alpha}$
P3	T + unary oracle	reflection on explicit trees	$f_{\varepsilon_0\cdot\alpha}$
P4	T + Brouwer trees + oracle	ramified reflection	$f_{\mathrm{BH}\cdot\alpha}$
P5	T + W-types at all finite types	two diagonals	$f_{\psi(\Omega_\omega)+\omega^2}$

P5 has by far the strongest base and the weakest amplifier, which suggests combining it with the others' amplifiers.

5 Contender 6

5.1 The object language

The language is P5's. Types are generated by $A, B ::= \mathbb{N} \mid A \to B \mid W\,A$, where $W\,A$ is the type of well-founded trees branching over A. Terms are raw (ill-typed terms default), with natural recursion

and a W-recursor into every type. Because A ranges over all types, the language has tree classes of every finite level, $W\mathbb{N}$, $W(W\mathbb{N})$, ..., and its calibration is $|L| \approx \psi(\Omega_\omega)$.

5.2 Oracles for free

P5's interpreter resolves variables by oldest-first levels. Evaluating a program in the one-element environment $[\langle \mathbb{N} \to \mathbb{N} \to \mathbb{N}, fs \rangle]$ therefore makes `var 0` denote the oracle family fs everywhere, even under binders. No new syntax is needed, and the smoke test confirms it: `var0 3 4` evaluates to $fs(3,4)$, and so does `(fun y => var0 y y) 5` with the family applied at $(5,5)$.

5.3 Reflection along the language's own trees

```
def reflect (fs : Family) (n : ℕ) : ℕ :=
  n + 1 + max (prefixMax (fun i => fs i n) n)
              (maximum (evalWith fs) (allTerms n))
def stage (seed : Family) : WTree ℕ → ℕ → ℕ
  | .leaf   => reflect seed
  | .node f => reflect (fun i => stage seed (f i))
def treesOfDepth (k : ℕ) : WTree ℕ :=
  .node fun i => defTree (nthD (allTerms k) i .zero)
def index : WTree ℕ := .node treesOfDepth
def contender_6 : ℕ := stage p5Family index 42
```

At a node the oracle family *is* the family of child stages, so a program may call any child at a computed index (P2's forest reflection). The new ingredient is `index`: its k-th child joins every $W\mathbb{N}$ tree denoted by a program of depth $\leq k$. The hierarchy therefore runs along the language's own ordinals, all of $|L|$, instead of an explicit notation system below BH. By the multiplication heuristic, Contender 6 is about $f_{|L|\cdot|L|}(42)$.

5.4 What is proved

- `contender_5_lt_contender_6`: the exact port of Contender 5 is below Contender 6.
- `p5_lt_contender_6` and `p5Family_lt_finalFamily`: P5 is beaten at every input.
- `defined_tree_eventually_lt`: every tree the language defines is eventually dominated.
- `program_lt_finalFamily`: every program of depth $\leq n$, run with the final oracles, is exceeded.

All of them depend only on `propext`. The definitions check in about 5 s and the proofs in about 6 s. A machine-checked audit shows that the definition never mentions `contender_5` or its winner selection. It also reports, candidly, that P5's seed contains the old language's maximum as a function of depth.

6 Engineering lessons

- **Toolchain drift changes axiom lists.** In Lean 4.34 a wildcard match arm compiles to a `_sparseCasesOn` helper built on `Nat.land`. `Nat.bitwise`'s termination proof uses `propext`, so "axiom-free on 4.19" did not survive the upgrade. The same happens to core `xs[i]?`. Exhaustive matches avoid it.

- **Tactics carry axioms.** `omega`'s certificates use `funext`, and so `Quot.sound`. Two calls were replaced by explicit lemma chains.
- **Never let the unifier normalize a contender.** Writing `iterate f 41 (f 42)` against `iterate f 42 42` made Lean try to evaluate the number. Generalizing the statement first, so that only $42 = 41 + 1$ is checked, fixes it.
- **Prove inequalities by inclusion, not monotonicity.** Reflection is not monotone in its oracle, because a program may test oracle values. Every comparison therefore exhibits an explicit occurrence: a visible oracle index or an enumerated program.

7 Contender 7: impredicative polymorphism

Contender 6 lives below the next collapsing level: everything it does is predicative. The next qualitative jump is *impredicative polymorphism*. The numeric functions definable in Girard's System F on Church numerals are exactly those provably total in second-order arithmetic Z_2, and $\psi(\Omega_\omega)$ is merely the ordinal of Π^1_1-CA_0, a small fragment of Z_2.

7.1 Why Lean can host System F

System F has no set-theoretic model in Lean's predicative `Type`: a model would have to interpret $\forall X.A$ as a product over *all* types, including itself. Girard's normalization proof needs much less. It interprets types as *reducibility candidates*, predicates on terms, and quantifies over all candidates. In Lean a candidate is a structure whose main field is $P : \mathrm{Tm} \to \mathrm{Prop}$. A statement "$\forall C : \mathrm{Cand}, \ldots$" is again a proposition, because `Prop` is impredicative. The normalization proof therefore goes through, even though there is no semantic model.

7.2 The formalization

1. **Untyped core** (`SystemF/Syntax`, `Reduction`): de Bruijn terms with an oracle constant `orc`; parallel substitution with its composition algebra; full β-reduction plus the δ-rule $\mathtt{orc}\,\overline{n} \to \overline{f(n)}$ on canonical Church numerals; SN as accessibility.
2. **Types** (`Types`): de Bruijn System F types and Curry-style typing of *erased* terms. A flag enables the oracle rule; numerals are typed without it.
3. **Candidates** (`Candidates`): CR1–CR3, where *neutral* means neither a λ nor the oracle (`orc a` can reduce at the head); the arrow construction; the intersection $\forall$; the interpretation $[\![A]\!]_\rho$; its renaming and substitution lemmas, proved as *equalities* of candidates via funext, propext and proof irrelevance; β-closure for λ; and adequacy. The oracle is justified afterwards. Numerals lie in every interpretation of `nat` (adequacy without the oracle rule), so the δ-reduct of $\mathtt{orc}\,a$ stays inside the candidate.

Theorem 2 (`F.sn_of_typed`)**.** *Every term typable in System F with the oracle is strongly normalizing under $\beta\delta$, for every oracle function f. The proof uses only* `propext` *and* `Quot.sound`*.*

7.3 From programs to numbers

Programs are Church-style terms (annotated λ, explicit type abstraction and application), so type inference is decidable. `infer_sound` states that inference implies the Curry typing of the erased term.

A program's value is computed as follows: check that it is closed of type $\texttt{nat} = \forall X.(X \to X) \to X \to X$; erase it; normalize it by the normal-order step function `nstep`, iterated by well-founded recursion on the SN certificate (the compiler runs this as an ordinary loop); and read the Church numeral. A smoke test shows the pipeline computing: $2^3 \mapsto 8$ and $3^{2^2} \mapsto 81$ by polymorphic instantiation, oracle calls $\texttt{orc}\,\overline{4} \mapsto 40$ for $f(n) = 10n$, and 0 for ill-typed programs.

7.4 The contender

```
def reflectF (f : ℕ → ℕ) (n : ℕ) : ℕ :=
  n + 1 + max (f n) (maximum (fvalue f) (ctmsUpTo n))
def F7 (n : ℕ) : ℕ := reflectF finalFamily n
def contender_7 : ℕ := F7 (F7 42)
```

The oracle is Contender 6's family. System F programs can call it and amplify it with second-order strength. The outer `F7` runs System F at a syntax budget already larger than Contender 6. Proved: `contender_6_lt_contender_7`, `finalFamily_lt_F7` (at every input), and `fvalue_lt_contender_7` (every System F program of depth $\leq$ `F7 42` is exceeded). The audit confirms that the constant `contender_6` itself never occurs in the definition; only Contender 6's family occurs, as the oracle.

Size. By the heuristics of Section 2, Contender 7 is about $f_{|\mathsf{Z}_2|}$ applied at an input beyond Contender 6. Here $|\mathsf{Z}_2|$ is the proof-theoretic ordinal of second-order arithmetic. Nobody has computed it, but it lies far beyond every ordinal named in this article.

8 Contender 8: type operators

System F_ω adds *type operators*. Kinds are $\kappa ::= \star \mid \kappa \to \kappa$; type constructors are simply typed λ-terms over kinds, with $\to$ and $\forall\kappa$ as constants of kind $\star$. Programs may now abstract over type operators. A typical example is $\Lambda F{:}\star \to \star.\, \lambda x{:}F\,\texttt{nat}.\, x$. The numeric functions definable in F_ω are those provably total in higher-order arithmetic, beyond second-order arithmetic.

8.1 Candidates over kinds

The semantic domain is defined by recursion on kinds: $\mathrm{Sem}(\star) = \mathrm{Cand}$ and $\mathrm{Sem}(\kappa_1 \to \kappa_2) = \mathrm{Sem}(\kappa_1) \to \mathrm{Sem}(\kappa_2)$, with the arrow on the right being an ordinary Lean function type. A type operator therefore *denotes a function on candidates*, and type-level β-reduction is literally function application. The interpretation recurses on syntax at an *expected kind*, returning a default element on kind mismatch. This is the same dynamic-cast idea as the raw interpreters of Contenders 5 and 6, and it keeps the definition structural. Type application records its argument's kind for this purpose. The substitution lemma, the instantiation lemma and the invariance of the interpretation under β-conversion are proved for well-kinded types. The beta case of invariance is one line: function application plus the instantiation lemma.

Theorem 3 (`Fw.sn_of_typed`)**.** *Every term typable in System F_ω with the oracle is strongly normalizing, using only* ***`propext`*** *and* ***`Quot.sound`****.*

8.2 Deciding type equality, soundly

Church-style checking must compare types up to β. Instead of proving a second normalization theorem for the type level, the checker compares *fuel-bounded* normal forms. Every normal-order step is a genuine β-step, so equal results witness joinability, and `Fw.infer_sound` inserts the corresponding conversions. A program needing more fuel is merely rejected. That loses programs but never admits an ill-typed one, and Contender 8 uses the syntax budget as the fuel. The smoke test checks $(\lambda x{:}(\lambda X{:}\star.X)\,\mathtt{nat}.\,x)\,\overline{3} \mapsto 3$ (rejected with fuel 0), and quantification over an operator, $(\Lambda F{:}\star\to\star.\lambda x{:}F\,\mathtt{nat}.x)\,[\lambda X.X]\,\overline{5} \mapsto 5$.

8.3 The contender

```
def family7 (n : ℕ) : ℕ := F7 (F7 n)       -- family7 42 = contender_7
def F8 (n : ℕ) : ℕ := reflectFw family7 n
def contender_8 : ℕ := F8 (F8 42)
```

A first attempt used `F7` itself as the oracle and tried to compare `F7 (F7 42)` with `F8 (F8 42)`. That needs monotonicity of reflection, which fails. Taking Contender 7's whole construction as the family makes the comparison immediate: `family7 42 < F8 42 < F8 (F8 42)`.

9 Contender 9: reflecting F_ω along the W-language's trees

9.1 Why not the Calculus of Constructions

The natural next rung, the Calculus of Constructions, turns out *not* to be a growth-rate jump. In pure CoC, term dependencies can be erased into F_ω (Geuvers and Nederhof), so the numeric functions CoC represents on Church numerals are essentially those of F_ω. This is why Loader's number sits in the F_ω/CoC class, and Contender 8 already lives there. A genuine jump beyond higher-order arithmetic requires *runtime-to-type dependency with large elimination*: types computed by recursion on input numbers, so that a single program uses types of unbounded order. That is the realm of the Calculus of Inductive Constructions, whose normalization needs a realizability model with universes. It is a much larger project, described in the roadmap.

9.2 A multiplicative step

While that project is pending, Contender 6's amplifier applies to any total language with an oracle. Contender 9 reflects F_ω along Contender 6's self-referential index, the join of every tree the W-language defines:

```
def stage9 (seed : Family) : WTree ℕ → ℕ → ℕ
  | .leaf   => reflect9 seed
  | .node g => reflect9 (fun i => stage9 seed (g i))
def F9 (n : ℕ) : ℕ := stage9 (fun _ => family8) index n
def contender_9 : ℕ := F9 (F9 42)       -- family8 n = F8 (F8 n)
```

`reflect9 fs n` exceeds the visible oracle values $fs(i,n)$, $i \le n$, and every F_ω program of depth $\le n$. F_ω's oracle is unary, so the family is packed as `orc` $(2^i(2m+1)) = fs(i,m)$. Programs can compute these codes and so call any child at any input; the theorem `unpair2_code` shows that the decoder inverts the code. The index trees must still come from the W-language: F_ω has no set-theoretic model, so its programs cannot hand trees back to Lean. By the multiplication heuristic, Contender 9

is about $f_{|\mathsf{Z}_\omega|\cdot\psi(\Omega_\omega)^2}$. Proved: `contender_8_lt_contender_9`, `family8_lt_F9` (at every input) and `defined_tree_eventually_lt9`.

9.3 Budgets for a growing stack

Each contender seeds itself with its predecessor's family, so its definition transitively imports the whole history. The full chain took 16–30 s under parallel load on the reference laptop. A fixed budget for the whole stack would eventually be exhausted by history alone. The rules now treat earlier accepted contenders like the precompiled core library: the 15 s/60 s budgets apply to the new contender. Contender 9's definition checks in about 2 s and its proofs in about 3 s, and the full clean build of all contenders takes about 28 s.

10 Contender 10: kind polymorphism

10.1 Unbounded orders without dependent types

Each F_ω program uses finitely many kinds, fixed in its text, and is provably total in higher-order arithmetic of some finite order. To get past that, a single program must reach kinds of *unbounded* order, chosen at run time. Full dependent types with large elimination would do it. A much cheaper mechanism suffices: types that quantify over *kinds*, $\forall\kappa.A$ (of kind $\star$). A step function

$$g : \ \forall\kappa.\ B(\kappa) \to B(\kappa \to \star),$$

iterated m times through the impredicatively encoded existential $\exists\kappa.B(\kappa) = \forall Y{:}\star\,.\,(\forall\kappa.\,B(\kappa) \to Y) \to Y$, climbs m orders, where m is a run-time number. A totality proof for such programs needs reducibility candidates for all finite types uniformly, which is beyond higher-order arithmetic.

Consistency. In pure type system terms this adds the rule $(\triangle, \star)$, types quantifying over the sort of kinds, but not $(\triangle, \square)$, kinds quantifying over kinds. The latter would give Girard's System U^-, which is inconsistent by Hurkens' paradox. With only $(\triangle, \star)$, kinds have no binders, so kind variables always range over closed kinds, a countable set. The Lean proof below is semantic, so it certifies normalization, and with it consistency.

10.2 The model

A *kind valuation* θ maps kind variables to closed F_ω kinds, and kind annotations are evaluated under θ. The single new semantic ingredient is

$$[\![\forall\kappa.A]\!]_{\theta,\rho} = \bigcap_{K \text{ closed}} [\![A]\!]_{K\cdot\theta,\rho},$$

a countable intersection of candidates. The renaming and substitution lemmas for kind variables hold unconditionally. Adequacy gains two cases: kind generalization (a fresh kind extends θ, and contexts are kind-shifted) and kind instantiation (choose the evaluated kind).

Theorem 4 (`Fk.sn_of_typed`)**.** *Every term typable in kind-polymorphic System F_ω with the oracle is strongly normalizing, using only* `propext` *and* `Quot.sound`*.*

The Church-style checker extends Contender 8's with kind abstraction and application, again proved sound. A smoke test types $\Lambda\kappa.\Lambda F{:}\kappa.\lambda x{:}\mathtt{nat}.x$ as $\forall\kappa.\forall F{:}\kappa.\,\mathtt{nat} \to \mathtt{nat}$. It runs it at kind $\star$ (giving 4) and at kind $\star \to \star$ with the identity operator (giving 6), and rejects an ill-kinded instantiation.

10.3 The contender

```
def family9 (n : ℕ) : ℕ := F9 (F9 n)      -- family9 42 = contender_9
def F10 (n : ℕ) : ℕ := reflectFk family9 n
def contender_10 : ℕ := F10 (F10 42)
```

Proved: `contender_9_lt_contender_10`, `family9_lt_F10` (at every input), and `fvalue_lt_contender_10`. The new definition modules, four for the language plus the contender, check in about 12 s.

11 Contender 11: reflecting the kind-polymorphic language

Contender 11 applies Contender 9's amplifier to Contender 10's language: forest reflection of kind-polymorphic F_ω along Contender 6's index, with oracle families packed into the unary oracle as before.

```
def F11 (n : ℕ) : ℕ := stage11 (fun _ => family10) index n
def contender_11 : ℕ := F11 (F11 42)      -- family10 n = F10 (F10 n)
```

As with Contender 9, the gain is multiplicative, of about $|\mathrm{F}^\kappa_\omega| \cdot \psi(\Omega_\omega)^2$, where F^κ_ω denotes Contender 10's kind-polymorphic language.

12 A second batch of proposals, and Contender 12

12.1 Five more proposals

Five more archives arrived, all proposing a "Contender 7" on top of this repository's Contender 6, written before Contenders 7–11 existed. All five definitions compile under Lean 4.34. Two proposals compile completely: P6 (operator reflection), whose author had not compiled it, and P7 (critical-point reflection). The other three (P8 ramified worlds, P9 model ramification, P10 semantic mirror) fail in their proofs.

All five amplify the W-language, of strength about $\psi(\Omega_\omega)$. P6 lifts reflection to *transformers* of families and iterates the lift, which exponentiates the index height. P7 stacks explicit Veblen-style critical-point notations above the language-defined index. P8–P10 iterate world or model constructions in which a lower world's whole hierarchy is exposed to the next world's programs. The resulting ordinals range from $|L|^{|L|}$ to Γ-type closures above $|L|$, all far below second-order arithmetic. Informally, the whole batch is vastly smaller than Contender 7 (System F), let alone Contenders 8–11.

12.2 Reusing the generic ideas

Two of the ideas are generic and compose with stronger languages. P6's operator stage needs only that its base transformer be *progressive*: every visible seed value lies below every output. P7's notations apply above any Lean tree. Contender 12 adopts both, with attribution in the source, over the kind-polymorphic F_ω reflection:

```
def fkTransformer : Transformer := fun fs _ n => reflect11 fs n
def tree12 := realize (hyper 42 (encode index))   -- P7 notations
def F12 (n : ℕ) : ℕ :=
  operatorStage fkTransformer tree12 (fun _ => family11) n n   -- P6
def contender_12 : ℕ := F12 (F12 42)
```

The base is progressive by one line (`oracle_lt_reflect11`), and P6's theorem `operatorStage_progressive` then gives `family11_lt_F12` at every input, with no structural facts about `tree12` required. The gain over Contender 11 is again inside the language's ordinal: the index factor grows from $\psi(\Omega_\omega)^2$ to roughly $|L|^{\Gamma\text{-closure above } |L|}$. The world and model constructions of P8–P10 need an object language that manipulates trees as data, which the impredicative languages lack. They are recorded for a future tree-capable language.

13 Contender 13: a predicative universe of operators

13.1 The idea

Kind polymorphism (Contender 10) lets one program reach every *finite* order uniformly. The next step is an object that collects all finite orders at once. Letting kinds quantify over kinds would do it impredicatively, but that is Girard's System U^-, which is inconsistent. The predicative alternative is a new kind `dyn` whose semantic domain is the dependent sum

$$\mathrm{Sem}(\texttt{dyn}) \;=\; \textstyle\sum_{k\ \text{dyn-free}} \mathrm{Sem}(k),$$

which is exactly the type `SVal` of Contender 8's model. It is a universe of all finite-order operators that does not contain itself. Two type-level constructs reach it: $\texttt{pack}\,K\,A : \texttt{dyn}$ for $A : K$, and $\texttt{unpack}\,K\,X : K$ for $X : \texttt{dyn}$, with the conversion $\texttt{unpack}\,K\,(\texttt{pack}\,K\,A) \to A$. Both are restricted to *dyn-free* kinds, a syntactic condition.

Kind variables now come in two sorts. Level-0 variables range over dyn-free kinds and can appear in `pack`/`unpack`. Level-1 variables range over *all* kinds, including `dyn`, $\texttt{dyn} \to \star$, and so on. The existential climbing trick of Contender 10, applied to level-1 variables, therefore reaches orders $\omega + n$ for run-time n.

13.2 The model

Semantic kinds are closed syntactic kinds, which have decidable equality, so the dynamic read `SVal.get` of Contender 6 still works. A decoding `toFw` sends dyn-free closed kinds to F_ω kinds, with $\mathrm{Sem}(K) = \mathrm{Sem}_{F_\omega}(k)$ whenever $\texttt{toFw}\,K = \texttt{some}\,k$. `pack` is `SVal.mk` through that cast, and `unpack` is `SVal.get` through the reverse cast. The lemma `unpack_pack` states that the read-back of a packed value is the value itself, which is what makes the new conversion sound. Level-0 kind variables are evaluated to F_ω kinds and level-1 variables to arbitrary kinds; $\forall\kappa_0$ and $\forall\kappa_1$ are intersections over the respective ranges.

Theorem 5 (`Fd.sn_of_typed`)**.** *Every term typable in dynamic-kind System F_ω with the oracle is strongly normalizing, using only* `propext` *and* `Quot.sound`*.*

The smoke test exercises all of it. A program typable only through $\texttt{unpack}\,\star\,(\texttt{pack}\,\star\,\texttt{nat}) \equiv \texttt{nat}$ evaluates to 3. Level-1 polymorphism instantiated at `dyn` evaluates to 5. Instantiating a level-0 variable with `dyn`, and packing at a kind containing `dyn`, are both rejected.

13.3 The contender

```
def family12 (n : ℕ) : ℕ := F12 (F12 n)    -- family12 42 = contender_12
def F13 (n : ℕ) : ℕ := reflectFd family12 n
def contender_13 : ℕ := F13 (F13 42)
```

Proved: `contender_12_lt_contender_13`, `family12_lt_F13` at every input, and `fvalue_lt_contender_13`. The new definition modules check in about 7.5 s.

13.4 A third batch of proposals

Six more proposals (P11–P16) arrived, written against Contender 10 (and one against Contender 6). Three compile completely (P11, P14, P15). The other three fail in their proofs, all through the same trap: a proof step that makes Lean compare or unfold a term containing a contender-sized number. P13's proof module did not terminate within minutes. Their amplifiers (self-fed iteration counts, crowns above the index, finite diagonal ranks of the autonomous lift) each add about a factor of ω to the index ordinal. Contender 12's tree-indexed operator stage subsumes them, and Contender 13's language exceeds them. P15 independently rediscovered Contender 11's exact construction.

14 Contender 14: the ladder of universes

14.1 The idea

`dyn` collects the operators of every dyn-free kind, but it never contains itself, and nothing collects operators whose kinds mention `dyn`. The predicative answer is a ladder. The level-ℓ language has atomic kinds $\texttt{dyn}_0, \ldots, \texttt{dyn}_{\ell-1}$, and atom $\texttt{dyn}_i$ denotes the `dyn` of the level-i language. Two type-level constructs move operators into and out of an atom:

$$\frac{A : K \qquad K \text{ fits } a}{\texttt{apack}\, a\, K\, A : a} \qquad \frac{X : a \qquad K \text{ fits } a}{\texttt{aunpack}\, a\, K\, X : K} \qquad \texttt{aunpack}\, a\, K\, (\texttt{apack}\, a\, K\, A) \;\longrightarrow\; A,$$

where K *fits* a when K is built from $\star$, $\to$ and atoms strictly older than a. The top-level `dyn` and both sorts of kind variables stay as in Contender 13. So a program can pack an operator of kind $\texttt{dyn}_0 \to \star$ into $\texttt{dyn}_1$ and read it back.

14.2 One normalization proof for every level

Writing a new model for each level would never finish, so the language and its model are *parametric*. The syntax (`SystemFL`) is parametric in a type α of atoms with decidable equality and an order (`AtomOrder`). The model takes a `Universe`: a domain $\mathrm{El}(a)$ for each atom, a packing map from the base kinds that fit a, a reading map, and the law

$$\mathrm{rd}_a\, k\, (\mathrm{pk}_a\, k\, v) = v.$$

Base kinds are $\star$, $\to$ and atoms, interpreted with $\mathrm{Sem}(a) = \mathrm{El}(a)$. `dyn` is the sum over base kinds, and `apack`/`aunpack` are pk/rd through the decoding cast. The read-back law is exactly what the new conversion needs.

Theorem 6 (`FL.sn_of_typed`)**.** *For every universe U, every term typable in the ladder language over U's atoms (with the oracle) is strongly normalizing.*

The ladder itself (`SystemFL/Ladder.lean`) instantiates this theorem at every level:

- atoms are $\texttt{Atom}\, 0 = \texttt{Empty}$ and $\texttt{Atom}\, (\ell + 1) = \texttt{Option}\, (\texttt{Atom}\, \ell)$. The new atom `none` lies above all older atoms $\texttt{some}\, a$;
- `nextUniv` interprets the new atom as `BVal` of the previous universe, i.e. as its `dyn`, and keeps the older atoms;
- packing and reading go through `lower`, which recognizes the level-$(\ell + 1)$ base kinds that avoid the new atom and turns them into level-ℓ base kinds. Semantic domains agree along it, so the read-back law at level $\ell + 1$ reduces to the law at level ℓ.

One engineering point matters. pk receives a proof that the kind fits, so the lowered kind is extracted with `Option.get` rather than a dependent `match`. The law then needs no rewriting under a dependent match, the usual source of ill-typed motives.

14.3 The contender

```
def ladderUpTo (n : ℕ) := ctmsUpTo (Atom.all n) n      -- level-n programs
def reflectFL (f : ℕ → ℕ) (n : ℕ) : ℕ :=
  n + 1 + maxN (f n) (maximum (fvalue n f n) (ladderUpTo n))
def family13 (n : ℕ) : ℕ := F13 (F13 n)     -- family13 42 = contender_13
def F14 (n : ℕ) : ℕ := reflectFL family13 n
def contender_14 : ℕ := F14 (F14 42)
```

The value of a level-ℓ program is computed by the certified normalizer with the SN proof `FL.sn_of_typed` applied to the level-ℓ universe `Univ f`. The reflection diagonalizes *at the meta level*. Input n uses the level-n language, so no fixed rung keeps up with F_{14}. Proved: `contender_13_lt_contender_14`, `family13_lt_F14` at every input, and `fvalue_lt_contender_14`. The smoke test packs λ_:$\texttt{dyn}_0$.`nat` into $\texttt{dyn}_1$ and applies its read-back to a packed `nat`, which evaluates to 4. It also confirms that an atom cannot hold its own kind.

The first budget run hit the 15 s definition limit (16.0 s on a loaded machine). Profiling showed the kind checker's soundness proof taking 2.4 s, with 6×6 case splits inside nested matches. Factoring the checker into small result functions (`arrK`, `appK`, `apackK`, ...) with standalone inversion lemmas cut that module from 5.8 s to 3.2 s. The definitions now check in about 10 s.

15 A fourth batch of proposals, and Contender 15

15.1 Four proposals over Contender 13

Four more proposals (P17–P20) arrived while Contender 14 was being built. All of them target Contender 13. All keep its language, dynamic-kind F_ω, and make its reflection *family-aware*: programs receive a packed countable oracle family, and a prefix maximum over the visible coordinates makes the transformer progressive without any monotonicity. Three compile as submitted (P17, P19, and P20, whose package says Lean was never run). P18 fails at a single proof step. A `simpa` misses the definitional equality `operatorStage` b `leaf` = `liftTransformer` b, and replacing it with `exact` repairs it. All audits pass with `propext` and `Quot.sound`.

Writing τ for the autonomous schedule above `tree12`, P18 and P19 stack operator stages into towers of height `family13` n or $n+1$, about $|F_d| \cdot \tau^\omega$. P17 and P20 independently found something stronger, a *rank hierarchy of transformer lifts*:

$$\mathrm{lift}_0 = \texttt{liftTransformer}, \qquad \mathrm{lift}_{k+1}(b) = \mathrm{stage}_{\mathrm{lift}_k}(\mathrm{lift}_k(b),\ \sigma),$$

where $\mathrm{stage}_L(b, \sigma)$ traverses the schedule σ, applying the lift L at every node. Rank $k+1$ uses the whole rank-k operator at every node, so each rank roughly exponentiates the index. Diagonalized at rank `family13` $n+1$, this gives about $|F_d| \cdot \varepsilon(\tau)$. It strictly generalizes Contender 12's single operator stage. All four exceed Contender 13 (each proves it), but by the base-language heuristic all four lie below the ladder of Contender 14.

15.2 Contender 15

The two routes compose. The lift hierarchy is generic over transformers, so it can be applied to the ladder's reflection instead of F_d's. Contender 15 adopts P17's modules unchanged, with attribution, and defines:

```
def reflectFLFamily (fs : Family) (n : ℕ) : ℕ :=
  n + 1 + maxN (prefixMax (fun i => fs i n) n)
    (maximum (FL.fvalue n (packFamily fs) n) (FL.ladderUpTo n))
def family14 (n : ℕ) : ℕ := F14 (F14 n)      -- family14 42 = contender_14
def schedule15 (n : ℕ) := realize (hyper (family14 n + 42) (encode tree12))
def F15 (n : ℕ) : ℕ :=
  diagonalLift (schedule15 n) (family14 n + 1) flFamilyTransformer
    (fun _ => family14) n n
def contender_15 : ℕ := F15 (F15 42)
```

Every proof follows an explicit occurrence: of the seed in the prefix maximum, of a child at coordinate 0 of a splice, or of rank k inside rank $k+1$. Proved: `contender_14_lt_contender_15`, `family14_lt_F15` at every input, and `fvalue_lt_F15` (every level-n ladder program of depth $\le n$ is exceeded). The definitions check in 2.9 s and the proofs in 3.1 s. Informally the size is about $|\mathrm{FL}_n| \cdot \varepsilon(\tau)$, diagonalized over the ladder level n.

16 Contender 16: a transfinite ladder

16.1 The length of the ladder is the lever

Contender 14 diagonalizes over finitely many universes. Letting a program choose its level at run time would add only about one more factor of ω. A much larger lever is the *length* of the ladder. The ladder language was parametric in its atoms from the start, and only the `Option` tower of Contender 14 was finite. So the question becomes: for which atom orders can Lean build a universe?

Theorem 7 (`FL.wfUniv`)**.** *For every type of atoms whose decidable order is well-founded, there is a universe. Hence (`FL.sn_of_typed`) the ladder language over those atoms is strongly normalizing.*

The domain of atom a is defined by well-founded recursion (`WellFounded.fix`):

$$\mathrm{El}(a) \;=\; \textstyle\sum_{k \text{ fits } a} \mathrm{Sem}(k), \qquad \mathrm{Sem}(b) = \mathrm{El}(b) \text{ for atoms } b < a.$$

A value stores its kind together with a proof that the kind fits. Reading compares kinds, which is decidable, so the law $\mathrm{rd}(\mathrm{pk}\, v) = v$ holds. The unfolding equation `WellFounded.fix_eq` is not definitional, so it is used through casts. Because types are erased, `WellFounded.fix` at `Type` compiles without a `noncomputable` marker. The universe at a limit atom such as ω holds operators from *every* lower level at once.

16.2 Well-founded Cantor normal forms

A non-increasing list $[a_1, \ldots, a_k]$ denotes $\omega^{a_1} + \cdots + \omega^{a_k}$, and the lexicographic order on such lists is the ordinal order. `Ordinal.wf_sorted` proves this order well-founded whenever the exponents are accessible. The proof uses neither transitivity nor trichotomy; asymmetry comes from accessibility. A sorted list headed by h splits into a run h^{k+1} and a tail below h. Two such lists compare like the pairs (k, tail), which gives an induction on the head, then on the run length, then on the tail. With natural-number exponents this gives notations below ω^ω (`OmegaPow`). Applied hereditarily it will give ε_0.

16.3 The contender

```
def transfiniteUpTo (n : ℕ) := ctmsUpTo (omegaPowUpTo n) n   -- atoms: ≤ n terms, exponents < n
def family15 (n : ℕ) : ℕ := F15 (F15 n)
def F16 (n : ℕ) : ℕ :=
  diagonalLift (schedule16 n) (family15 n + 1) tTransformer (fun _ => family15) n n
def contender_16 : ℕ := F16 (F16 42)
```

`tTransformer` is family-aware reflection over the transfinite ladder, normalized by `sn_of_typed (wfUniv f omegaPow_wf)`. At depth n, programs reach levels up to $\omega^{n-1}\cdot n$. Proved: `contender_15_lt_contender_16`, `family15_lt_F16` and `tvalue_lt_F16`. The smoke test packs an operator of kind $\mathtt{dyn}_3 \to \star$ into the level-ω universe and evaluates it to 4. The opposite direction is rejected. The definitions check in 4.8 s and the proofs in 2.8 s.

17 Contender 17: a ladder of length ε_0

Notations below ε_0 are trees $\mathtt{node}\,[a_1, \ldots, a_k]$ denoting $\omega^{a_1} + \cdots + \omega^{a_k}$. They are compared lexicographically, child by child, and a tree is normal when its children are normal and non-increasing. On arbitrary trees the lexicographic order is *not* well-founded. For example, $t_{k+1} = \mathtt{node}\,[0, t_k]$ descends forever starting from $t_1 = \omega$. So the sorted-list lemma cannot be applied to all trees.

The fix is to build normality into the Boolean element order, $\mathrm{lt}_P(a, b) := \mathrm{nf}(a) \wedge \mathrm{lt}(a, b)$, and apply the generic lemma *to that order*. Two short lemmas then finish the proof:

- `acc_of_children`: if the list of children of t is accessible (sorted-lexicographically), then t is accessible. A smaller normal tree has a smaller sorted list of children;
- `acc_nf`: every normal tree is accessible. By structural recursion its *head* child is accessible, so `acc_head` makes the list of children accessible. The other children lie below the head and need no recursion.

`Ordinal.epsilon0_wf` transfers this to the subtype of normal forms, and `FL.wfUniv` supplies the universes unchanged. At depth n, programs use atoms of height $\leq n$, i.e. levels up to towers $\omega^{\omega^{\cdots}}$ of height about n. The contender has the same shape as before, `F17 n := diagonalLift (schedule17 n) (family16 n + 1) eTransformer (fun _ => family16) n n`. Proved: `contender_16_lt_contender_17`, `family16_lt_F17` and `evalue_lt_F17`. The smoke test packs an operator over level ω into the level-ω^ω universe (value 5), and the opposite direction is rejected. The definitions check in 2.0 s and the proofs in 1.9 s.

18 Contender 18: a ladder of length Γ_0

18.1 Veblen terms and a case-split proof

Terms are sums $\mathtt{node}\,[\varphi(a_1, b_1), \ldots, \varphi(a_k, b_k)]$ of principal terms (a mutual, nested inductive). Sums are compared lexicographically, and principal terms by Veblen's rule. The rule is implemented as a Boolean function by mutual well-founded recursion on sizes:

```
def ltP : P → P → Bool
  | .phi a b, .phi c d =>
      if ltV a c then ltVP b c d          -- b < phi(c,d)
      else if decide (a = c) then ltV b d
      else leVP a b d                     -- phi(a,b) <= d
```

A term is normal when every sum is non-increasing, hereditarily. No fixed-point condition is imposed, so non-canonical terms such as $\varphi(0, \varepsilon_0)$ are allowed. They simply sit somewhere in the order. The genuine Veblen normal forms are among the normal terms, so the order type is at least Γ_0.

Theorem 8 (`Veblen.acc_phi`, `gamma0_wf`)**.** *If a and b are accessible, so is $\varphi(a, b)$. Hence every normal term is accessible, and the order on normal terms is well-founded.*

The proof is Schütte's, reorganized so that it only follows the *definitional* case split of `ltP`. It inducts on a, then on b, then on the size of a predecessor $\varphi(e, f)$:

- $e < a$ and $f < \varphi(a, b)$. Then f is 0 or headed by a smaller principal term below $\varphi(a, b)$, accessible by the size induction, so `acc_head` makes f accessible. The induction on a applies to e;
- $e = a$ and $f < b$: by the induction on b;
- otherwise $\varphi(e, f) \le b$: then the one-term sum $\varphi(e, f)$ is below or equal to b, hence accessible (`accP_of_le`).

Transitivity, trichotomy and ordinal arithmetic are never needed. Sums are handled by `SortedLex` as for ε_0.

One engineering note. The first version depended on `Classical.choice`. `omega` had been called on a goal of the form `AccP q`, after deriving an arithmetic contradiction. Because that goal is not decidable, `omega` used `Classical.byContradiction`. Writing `absurd hqs (by omega)` gives `omega` a decidable goal, and the axioms return to `propext` and `Quot.sound`.

18.2 The contender

`F18 n := diagonalLift (schedule18 n) (family17 n + 1) gTransformer (fun _ => family17) n n`, with `family17` $n = F_{17}(F_{17}\, n)$ and reflection over the ladder whose atoms are `Gamma0`. Proved: `contender_17_lt_contender_18`, `family17_lt_F18` and `gvalue_lt_F18`. The smoke test packs an operator over level ε_0 into the level-$\varphi(\varepsilon_0, 0)$ universe (value 6). The definitions check in 2.4 s and the proofs in 1.8 s.

19 Contender 19: a ladder of large-Veblen length

19.1 Klammersymbole

Binary Veblen generalizes to φ applied to an *argument array*: a finite list of entries (position, value), sorted by decreasing position. In Schütte's Klammersymbole the positions are themselves terms. Terms are now a three-way mutual inductive (sums, principal terms $\varphi(A)$, entries). Arrays are compared lexicographically by entries, with entries ordered by position and then value. This is exactly the order "first difference at the most significant position". Principal terms follow the generalized rule:

$$\varphi(A) < \varphi(B) \iff \begin{cases} A < B \text{ and every value of } A \text{ is } < \varphi(B), \text{ or} \\ A \neq B,\ A \not< B, \text{ and } \varphi(A) \le \text{some value of } B. \end{cases}$$

Binary Veblen is the case of positions 0 and 1: $\varphi(1@1) = \varepsilon_0$ and $\varphi(1@2) = \Gamma_0$. With positions ranging over the notation itself, the normal forms reach the large Veblen ordinal.

19.2 Well-foundedness

The proof still follows only the comparison's case split. Two new ingredients are needed:

- `SortedLexQ`: the sorted-list lemma restricted to a Prop-valued predicate. An entry is *good* when it is normal and its position and value are accessible. Good entries are accessible (`accE`: induction on the position, then the value). Hence sorted arrays of good entries are accessible (`accQ_sorted`).
- `posAcc`: if $\Gamma < A$ as sorted arrays, every position of Γ lies below or at a position of A, so it is accessible when A's positions are.

`LVeblen.acc_phi` inducts on the array, then on the size of the predecessor $\varphi(\Gamma)$. In the first case the values of Γ are below $\varphi(A)$ and accessible by the size induction, and the positions by `posAcc`. So Γ is a smaller good array and the array induction applies. In the second case $\varphi(\Gamma)$ lies below a value of A. The whole module checks in about 2 s, and the axioms are `propext` and `Quot.sound`.

19.3 The contender

`F19 n := diagonalLift (schedule19 n) (family18 n + 1) lTransformer (fun _ => family18) n n`, with reflection over the ladder whose atoms are `LargeVeblen`. Proved: `contender_18_lt_contender_19`, `family18_lt_F19` and `lvalue_lt_F19`. The smoke test packs an operator over level ε_0 into the universe at the small Veblen ordinal $\varphi(1@\omega)$ (value 7). The definitions check in 4.3 s and the proofs in 1.8 s.

20 Contender 20: a ladder of Bachmann–Howard length

20.1 Collapsing, and why the case split no longer suffices

Buchholz-style terms with two levels are sums of principal terms $D_0 a$ ($\psi_0(a)$, countable) and $D_1 a$ ($\psi_1(a)$, at least Ω). Principal terms are compared lexicographically (level, then argument), and sums lexicographically, so every D_1 term lies above every D_0 term. This raw order is not well-founded:

$$D_1 0 \;>\; D_0 D_1 0 \;>\; D_0 D_0 D_1 0 \;>\; \cdots$$

Normal forms exclude this. $D_u a$ is normal only when every element of $G_u(a)$ is below a, where $G_u(D_v b) = \emptyset$ for $v < u$ and $\{b\} \cup G_u(b)$ otherwise. In the chain, $D_0 D_1 0$ is normal, but $D_0 D_0 D_1 0$ is not, because $D_1 0 \in G_0(D_0 D_1 0)$ is not below $D_0 D_1 0$. The normal forms reach $\psi_0(\varepsilon_{\Omega+1})$, the Bachmann–Howard ordinal.

20.2 The impredicative argument in Lean

Let M be the set of terms whose *top-level D_0-components* (reached through sums and D_1 only) are accessible.

1. `accM_V` (Claim 1): the order restricted to M is well-founded. This part is predicative. D_0 components are accessible by assumption, D_1 terms by induction on their argument, and sums by `SortedLexQ`.
2. `accD0` (Claim 2): if a is accessible within M, then $D_0 a$ is accessible. A predecessor $D_0 c$ has $c < a$. Its components $D_0 f$ satisfy $f < c$ by normality, hence $f < a$ by *transitivity*, and they are smaller, so an inner size induction puts c in M. The outer induction then applies to c.

3. Claim 3: every normal $D_0 c$ is accessible (size induction plus Claims 1 and 2). $D_1 b$ is accessible whenever b is, because its predecessors are D_0 terms or $D_1 c$ with $c < b$. Hence every normal term is accessible (`bachmannHoward_wf`).

This is the point where the well-foundedness proof itself becomes impredicative. Claim 1 is an accessibility statement *relative to* a set defined from accessibility, and Lean's inductive `Acc` carries it without any extra axiom. Transitivity, avoided until now, is needed in Claim 2. The order is purely lexicographic and structurally recursive, so transitivity is a short mutual induction. `SortedLexQ` gained `accQ_head_of_acc`, which starts from one accessible head rather than a global hypothesis. The module checks in about 2 s.

20.3 The contender

`F20 n := diagonalLift (schedule20 n) (family19 n + 1) bTransformer (fun _ => family19) n n`, with reflection over the ladder whose atoms are `BachmannHoward`. Proved: `contender_19_lt_contender_20`, `family19_lt_F20` and `bvalue_lt_F20`. The smoke test packs an operator over level $\psi_0(\Omega)$ into the universe at Ω (value 8). The definitions check in 2.9 s and the proofs in 1.9 s.

21 Contender 21: Buchholz ladders up to $\psi_0(\Omega_\omega)$

21.1 Why more levels are hard

With levels $0, 1, 2, \ldots$ the two-level argument of Contender 20 regresses. The order restricted to M_0 contains every $D_1 d$ with $d \in M_0$ arbitrary, and every $D_2 b$ lies above all of them. So Claim 1 at level 0 needs a level-1 collapse *relative to* M_0, which needs a level-2 collapse relative to a smaller scope, and so on. There is no bound, because high levels hide inside low arguments: $\psi_0(\Omega_9) < \Omega_1$. The literature resolves this with Buchholz's distinguished sets.

21.2 A level bound turns it into a downward induction

For a fixed bound L on all levels, the regress stops. A *scope* is a predicate Z on principal terms, and the order is restricted to normal in-scope terms. For $u \le L + 1$ let

> $R(u)$: if every in-scope term of level $< u$ is accessible and scopes are closed under arguments at levels $\ge u$, then every in-scope normal principal term is accessible.

$R(L + 1)$ holds because every level is below $L + 1$. `Buch.stepR` derives $R(u)$ from $R(u + 1)$:

1. Restrict the scope to Z', the in-scope terms whose top-level u-components (through sums and higher levels) are accessible. $R(u + 1)$ applied to Z' makes the Z'-order well-founded (Claim 1).
2. Induction along it, an inner size induction and transitivity make $D_u a$ accessible for every a accessible within Z' (Claim 2).
3. Every in-scope $D_u c$ is then accessible (Claim 3).
4. $R(u + 1)$ applied to Z itself gives the conclusion.

$R(0)$ for the full scope gives `buchholz_wf L`: the notation with levels $\le L$ is well-founded. Its normal forms (for $D_u a$: $f < a$ for every top-level u-component $D_u f$ of a) include Buchholz's, so the order type is at least $\psi_0(\varepsilon_{\Omega_L + 1})$. Relations are written as `LTQ (scope ∧ nf) ltP`, so `SortedLexQ` and `Subrelation.accessible` do all the bookkeeping between scopes.

21.3 The contender

As in Contender 14, the ladder is diagonalized at the meta level. Input n uses the atoms `Buchholz n`, with the universe `wfUniv f (buchholz_wf n)`. The ladder lengths therefore climb through $\psi_0(\varepsilon_{\Omega_n+1})$ towards Buchholz's ordinal $\psi_0(\Omega_\omega)$, the proof-theoretic ordinal of Π^1_1-CA$_0$. `F21 n := diagonalLift (schedule21 n) (family20 n + 1) zTransformer (fun _ => family20) n n`. Proved: `contender_20_lt_contender_21`, `family20_lt_F21` and `zvalue_lt_F21`. The smoke test (level bound 2) packs an operator over level $\psi_0(\Omega_2)$ into the universe at Ω_1 (value 9). The definitions check in 4.0 s and the proofs in 2.5 s.

22 Contender 22: a tower of relativized Buchholz notations

22.1 The barrier at $\psi_0(\Omega_\omega)$

Contender 21's argument needs a bound on the levels: the downward induction starts at $R(L+1)$. A single notation for all levels has no such start, and the relativized argument regresses forever. The literature crosses this barrier with Buchholz's *distinguished sets*: sets X such that, for each $\alpha \in X$, X agrees below the next regular above α with the well-founded part of a closure $C^\alpha(X)$. The maximal distinguished class is then shown to contain every notation. Arai's lecture notes present the technique for reflection notations. Porting it means formalizing a large part of the notation theory (closure sets, their monotonicity, and the comparison facts), so it remains on the roadmap.

22.2 Relativizing to a base

A cheaper step stays within finitely nested accessibility. The level-bounded systems are relativized to a type of *base atoms* with a transitive, well-founded order (`BuchholzBase.lean`). Atoms $B\,b$ lie below every collapse but may occur inside collapse arguments, so each collapse ranges over everything below it. The well-foundedness proof is Contender 21's `stepR`. Only the treatment of atoms changes: they are accessible in every scope by the base's own well-foundedness (`accB`), not by the level induction.

The union of all level-bounded systems over a base, ordered by the bound *first*, is again a transitive, well-founded, decidable order. It is well-founded by induction on the bound and then `acc_nf`. So it can serve as the base of the next round:

```
structure WFO where            -- order, transitivity, well-foundedness, enumeration
  T : Type
  ...
def tower : ℕ → WFO
  | 0 => WFO.empty
  | k + 1 => (tower k).next      -- all bounded systems over tower k
```

`tower 1` has order type $\psi_0(\Omega_\omega)$. Each further step collapses over everything below it, heuristically giving $\psi_0(\Omega_\omega \cdot k)$ for `tower k`.

22.3 The contender

Input n uses the ladder over `tower n`, with the universe `wfUniv f (tower_wf n)`. `F22 n := diagonalLift (schedule22 n) (family21 n + 1) twTransformer (fun _ => family21) n n`. Proved: `contender_21_lt_contender_22`, `family21_lt_F22` and `tvalue22_lt_F22`. The smoke test packs an operator over a base atom of `tower 2` into the universe at ψ_0 of that atom (value 10). The definitions check in 3.7 s and the proofs in 1.9 s.

23 Proposal P22 and Contender 23: transfinite ranks

Two more proposals arrived. P21 is a tower of operator stages over Contender 13's language. It compiles even though its package says it was never compiled, and it is subsumed by the rank hierarchy of lifts. P22 makes a real improvement to that hierarchy. The rank, a natural number in P17's version, becomes a *well-founded W-tree*:

```
def wDiagonalLift (schedule : WTree ℕ) : WTree ℕ → TransformerLift
  | .leaf => liftTransformer
  | .node f =>
      let lower := spliceLifts (fun i => wDiagonalLift schedule (f i))
      fun base => stageLift lower (lower base) schedule
```

At a node, the lifts of all children are spliced into one lift, which is used twice: to pre-amplify the base, and as the local lift along the schedule. The rank tree is an autonomous notation above the schedule, so the hierarchy has a large countable rank instead of ω. P22 compiles and passes the audit. It targets Contender 19's language, so it lies below Contenders 20–22. But the amplifier composes with any ladder.

Contender 23 adopts P22's modules unchanged, with attribution, and applies them to Contender 22's tower ladders:

```
def rank23 (n : ℕ) := realize (hyper (family22 n + 42) (encode (schedule23 n)))
def F23 (n : ℕ) : ℕ :=
  wDiagonalLift (schedule23 n) (rank23 n) twTransformer (fun _ => family22) n n
```

Proved: `contender_22_lt_contender_23`, `family22_lt_F23` and `tvalue22_lt_F23`. The definitions check in 2.6 s and the proofs in 3.9 s.

24 Contender 24: towers along well-founded indices

Contender 22's tower iterates "all level-bounded systems over the previous stage" along ℕ. Nothing forces the index to be ℕ. `TowerAlong.lean` builds the tower along any well-founded index J:

```
def towerAlong (J : WFO) (j : J.T) : WFO :=
  (sigma (sub J j) (fun i => towerAlong J i.1)).next   -- over all stages below j
termination_by j
def limitAlong (J : WFO) : WFO := sigma J (towerAlong J)
def roundTower : ℕ → WFO
  | 0 => tower 1
  | k + 1 => (roundTower k).limitAlong                  -- indexed by the last round
```

`sigma J F` is the union of a family of orders, ordered by index first. Its decidable equality, transitivity and well-foundedness (induction on the index, then on the element) are proved once and for all. The stage at j is heuristically of order type $\psi_0(\Omega_\omega \cdot (j+1))$. Since each round is indexed by the previous one, the rounds climb the fixed points of $x \mapsto \psi_0(\Omega_\omega \cdot x)$.

One implementation point. `WellFounded.fix` producing *data*, here records with order functions and enumerations, would need `noncomputable`, which the rules forbid. Recursion via `termination_by`, with a high-priority `WellFoundedRelation` instance built from J's well-foundedness, compiles normally. The smoke test computes 165 atoms of `roundTower 1` at depth 2 and runs a pack/unpack program between two of them.

Input n uses the ladder over `roundTower n`, with Contender 23's W-ranked lift hierarchy seeded with Contender 23's family. Proved: `contender_23_lt_contender_24`, `family23_lt_F24` and `rvalue_lt_F24`. The definitions check in 2.5 s and the proofs in 2.2 s.

25 Contender 25: well-foundedness for free

25.1 More proposals, and one decisive idea

Four more proposals arrived (P23–P26), all over Contender 19's language. P23 and P26 raise the lift hierarchy one type level (lifts of lifts). P24 rediscovers P22's W-tree-ranked hierarchy. They all compile, P23 after the same one-step `simpa` repair as P18, and none exceeds the current contenders. P25 contributes the decisive idea. Its atoms are *finite addresses in a computable W-tree*. Following an address is a computation, so validity is decidable. The order between addresses is well-founded *by structural recursion on the tree*, whatever ordinal the tree denotes.

All ladders since Contender 16 paid for their length with a hand-written well-foundedness proof, and that cost stopped the notations at Buchholz's ordinal. With addresses, the cost disappears. The ladder length is the rank of *any* tree that Lean can define, and Lean defines trees by structural recursion, with no proof obligations.

25.2 Addresses

`TreeAddress.lean` uses tree ordinals (zero, successor, countable limits, and suprema over two index types, used by collapsing) and the *strict descendant* order on valid addresses. P25 used immediate children. Accessibility of an address follows by recursion on the subtree it reaches (`acc_follow`). A strict descendant of a is a descendant of some child of a (`isDesc_split`), and addresses below a dead end are invalid. `addr_wf` uses only `propext`.

25.3 Collapsing functions by structural recursion

`TreeCollapse.lean` defines Buchholz-style collapsing on tree ordinals for all finite levels at once. Level-$(w{+}1)$ trees have suprema over all lower levels (lumped in one index type) and over level w. The argument space has suprema at every finite level. Then

```
def D : (v : ℕ) → TopT → LevT v
  | v, .zero => LevT.Omega v
  | v, .succ t => LevT.epsNext v (D v t)
  | v, .lim f => LevT.lim v fun n => D v (f n)
  | v, .supH g =>        -- below v: keep the supremum; for every j >= v: diagonalize
      ... D v (g (v + i, seq (v + i) n)) ...   -- seq j (n+1) = D j (g (j, seq j n))
```

Lean accepts this as structural recursion: every recursive call is on `g x` for some `x`. `D` depends on no axioms at all. The countable tree `bigTree n` collapses $n{+}1$ further ε-numbers above $\Omega_\omega + 1$, where $\Omega_\omega = \lim_k \Omega_k$. Heuristically its rank is $\psi_0(\varepsilon_{\Omega_\omega + n})$. That is beyond Buchholz's ordinal, and in exactly the range where a notation-based proof would need distinguished sets.

25.4 The contender

Input n uses the ladder over `Addr (bigTree n)`, with the universe `wfUniv f (addr_wf (bigTree n))` and the W-ranked lift hierarchy seeded with Contender 24's family. Proved: `contender_24_lt_contender_25`, `family24_lt_F25` and `avalue_lt_F25`. In the smoke test, an operator over the universe at address $(3, 5)$ packs into the root's universe (value 12). The definitions check in 3.5 s and the proofs in 2.2 s.

26 Contender 26: transfinitely many levels

Contender 25's collapsing functions have finitely many levels. With tree ordinals, the levels can themselves be countable tree ordinals (`TreeLevels.lean`). The type of everything below level a is defined by structural recursion on the level tree:

$$\mathrm{Low}(0) = \emptyset, \qquad \mathrm{Low}(a{+}1) = \mathrm{Low}(a) \oplus \mathrm{Lev}(a), \qquad \mathrm{Low}(\lim f) = \textstyle\sum_i \mathrm{Low}(f\,i) \oplus \mathrm{Lev}(f\,i),$$

with $\mathrm{Lev}(a)$ the trees with suprema over $\mathrm{Low}(a)$. Embedding a level into a higher one extends suprema by zero, so no retraction is needed.

A difficulty arises for arguments. Splitting a supremum into the parts below and above the target level would need a decidable order on levels, and tree orders are not decidable. The fix is to let suprema carry their level. The argument type has a constructor `supAt (c : CTree) (g : Low c → LTop)`, which is still strictly positive. Then

```
def DL : (a : CTree) → LTop → Lev a
  | a, .supAt c g =>
      match c, g with
      | .succ c', g => -- x0 = Omega c',  x(k+1) = DL c' (g x(k));  result lim_k DL a (g x(k))
      | .lim f, g   => -- the limit over i of the successor case at f i
      ...
```

Again every recursive call is on `g x`, so the definition is structural, and it uses only `propext`. `ladderTree n` iterates $\Lambda \mapsto \mathrm{DL}\,0\,(\varepsilon_{\Omega_\Lambda+1})$ $n{+}1$ times, starting from Contender 25's tree. Heuristically this climbs toward the fixed points of $\alpha \mapsto \psi_0(\Omega_\alpha)$. Input n uses the ladder over `Addr (ladderTree n)`, with the W-ranked lift hierarchy seeded with Contender 25's family. Proved: `contender_25_lt_contender_26`, `family25_lt_F26` and `lvalue26_lt_F26`. The definitions check in 4.4 s and the proofs in 4.9 s.

27 Contender 27: universes of level codes

In Contender 26 the levels are indexed by countable codes, so iterating $\Lambda \mapsto \psi_0(\Omega_\Lambda)$ stalls at a countable fixed point. To pass it, level codes must branch over *uncountable* level elements. Doing that in one step would be an inductive–recursive definition, which Lean does not have. The stratified version is available (`TreeUniverses.lean`):

```
structure Univ : Type 1 where
  Code : Type
  Low : Code → Type                          -- everything below level c
  sample : (c : Code) → ℕ → Option (Low c)   -- elements for diagonals
  top : Code
inductive NCode (U : Univ) : Type where      -- codes of the next universe
  | zero | succ (t : NCode U) | lim (f : ℕ → NCode U)
  | supAt (c : U.Code) (h : U.Low c → NCode U)
```

The next universe's `Low` is defined by structural recursion on `NCode U`, so its codes denote levels as large as the elements of the previous universe's levels. Iterating gives the tower `univ k`. Each universe has its own collapse `DN`, again by structural recursion. Successor codes diagonalize through the predecessor. Limit codes take limits. A code whose supremum ranges over an old `Low c` diagonalizes along sample elements of `Low c`. The samples are needed because these index types have no decidable equality, but the strength comes from the diagonal. `universeLadderTree n`

collapses $\varepsilon_{\Omega_{\text{top}}+1}$ in universe $n{+}1$ and uses no axioms at all. Heuristically the nesting $\Omega_{\Omega_{\cdots}}$ deepens with n, toward the first fixed point of $\alpha \mapsto \Omega_\alpha$, an inaccessible-like level.

Input n uses the ladder over `Addr (universeLadderTree n)`, with the W-ranked lift hierarchy seeded with Contender 26's family. Proved: `contender_26_lt_contender_27`, `family26_lt_F27` and `uvalue_lt_F27`. The definitions check in 2.8 s and the proofs in 2.5 s.

28 Contender 28: universes along tree ordinals

Contender 27 stacks universes of level codes along $\mathbb{N}$. If a universe carries a *sequence* of large codes instead of one, the union of a sequence of universes is again a universe, with codes $\sum_i (F\,i)$.Code. Universes can then be stacked along any countable tree ordinal, by structural recursion (`TreeUniverses2.lean`):

```
def univT : CTree → Univ
  | .zero => Univ.base
  | .succ a => (univT a).next
  | .lim f => Univ.union fun i => univT (f i)
```

The number of stacked universes can be the previous round's collapsed tree: `nextRound` collapses inside the universe after `univT` of the previous tree, and `roundTree n` iterates this $n{+}1$ times from Contender 27's tree. No axioms are used. Input n uses the ladder over `Addr (roundTree n)`, with the W-ranked lift hierarchy seeded with Contender 27's family. Proved: `contender_27_lt_contender_28`, `family27_lt_F28` and `rtvalue_lt_F28`. The definitions check in 2.2 s and the proofs in 2.4 s.

29 Roadmap

Longer ladders. The ladder now has whatever length Lean can prove well-founded; ε_0, Γ_0 and the large Veblen ordinal are done, the last by Klammersymbole, and so is the Bachmann–Howard ordinal by collapsing (Contender 20), and $\psi_0(\Omega_\omega)$ in the limit of bounded Buchholz systems (Contender 21). Contender 22 iterates bounded systems over a base. Contender 25 sidesteps notations entirely: tree ordinals defined by structural recursion need no well-foundedness proof. Contender 26 adds tree-indexed levels, and Contender 27 universes of level codes. Beyond that lie levels closed under codes for *all* universes at once, and Mahlo-like collapsing hierarchies, in which levels are closed under level-forming operations. All of them are definable by structural recursion in Lean's inductive types. Collapsing functions need genuine normal forms: in Buchholz's raw order, $D_1 0 > D_0 D_1 0 > D_0 D_0 D_1 0 > \cdots$ descends forever. Beyond that, collapsing-function notations ($\psi(\Omega_\nu)$ and further) are well-founded in Lean's type theory, and each one plugs into `wfUniv` unchanged.

Reflection along trees. As with Contenders 9 and 11, reflecting the ladder along the W-language's trees is a cheap multiplicative step.

Large elimination and trees. Runtime-to-type dependency with large elimination remains the route to a CIC-like system. A language able to hand trees back to Lean would unlock the world and model constructions of P8–P10.

30 Status

Contender	idea	informal size
5 (Coq)	System T, depth 42	$f_{<\varepsilon_0}(42)$
6 (Lean)	W-language, reflected along its own trees	$f_{\psi(\Omega_\omega)^2}(42)$
7 (Lean)	System F with Contender 6 as oracle	$f_{\lvert\mathsf{Z}_2\rvert}(>C_6)$
8 (Lean)	System F_ω with Contender 7 as oracle	$f_{\lvert\mathsf{Z}_\omega\rvert}(>C_7)$
9 (Lean)	F_ω reflected along W-trees	$f_{\lvert\mathsf{Z}_\omega\rvert\cdot\psi(\Omega_\omega)^2}(>C_8)$
10 (Lean)	kind-polymorphic F_ω	beyond higher-order arithmetic
11 (Lean)	10, reflected along W-trees	multiplicative gain over 10
12 (Lean)	10, operator-level reflection (P6, P7 ideas)	exponential index over 11
13 (Lean)	dynamic kind: a universe of operators	uniform orders $\omega + n$
14 (Lean)	ladder of universes, level n at input n	orders $\omega \cdot n$, diagonalized
15 (Lean)	14 with P17's rank hierarchy of lifts	$\lvert\mathrm{FL}_n\rvert \cdot \varepsilon(\tau)$
16 (Lean)	15 over a ladder of length ω^ω	levels $< \omega^n$ at input n
17 (Lean)	15 over a ladder of length ε_0	levels $< \omega_n$ (towers) at input n
18 (Lean)	15 over a ladder of length Γ_0	φ-towers of height n at input n
19 (Lean)	15 over a ladder of large-Veblen length	Klammersymbole of height n at input n
20 (Lean)	15 over a ladder of Bachmann–Howard length	ψ-terms of height n at input n
21 (Lean)	15 over Buchholz ladders, level bound n at input n	lengths up to $\psi_0(\Omega_\omega)$
22 (Lean)	15 over a tower of relativized Buchholz notations	beyond $\psi_0(\Omega_\omega)$
23 (Lean)	22 with P22's W-ranked lift hierarchy	transfinite rank over 22
24 (Lean)	towers along well-founded indices, self-indexed rounds	fixed points of $x \mapsto \psi_0(\Omega_\omega x)$
25 (Lean)	ladders over addresses in collapsed tree ordinals	$\psi_0(\varepsilon_{\Omega_\omega+n})$ at input n
26 (Lean)	collapsing with tree-indexed levels	toward fixed points of $\alpha \mapsto \psi_0(\Omega_\alpha)$
27 (Lean)	universes of level codes	toward the first fixed point of $\alpha \mapsto \Omega_\alpha$
28 (Lean)	universes stacked along collapsed trees	iterated universe heights